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SIMULATION EXPERIMENT – 3

Design of IIR Filters Using Impulse Invariance and Bilinear Transformation Methods Using MATLAB

MATLAB CODE

```
% Design of IIR Filters Using Impulse Invariance and Bilinear Transformation

clc;

clear;

close all;

% Sampling Frequency

Fs = 1000;

T = 1/Fs;

% Analog Filter Specifications

N = 4;          % Filter Order

Fc = 100;       % Cutoff Frequency (Hz)

Wc = 2*pi*Fc;     % Analog Cutoff Frequency (rad/s)

% Analog Butterworth Prototype

[b,a] = butter(N,Wc,'s');

%% Impulse Invariance Transformation

[b_imp,a_imp] = impinvar(b,a,Fs);

%% Bilinear Transformation (Without Prewarping)

[b_blt,a_blt] = bilinear(b,a,Fs);

%% Bilinear Transformation (With Prewarping)

Wp = 2*Fs*tan(pi*Fc/Fs);     % Prewarped Analog Frequency

[bp,ap] = butter(N,Wp,'s');

[b_pre,a_pre] = bilinear(bp,ap,Fs);

%% Frequency Responses

[H1,f] = freqz(b_imp,a_imp,1024,Fs);

[H2,~] = freqz(b_blt,a_blt,1024,Fs);

[H3,~] = freqz(b_pre,a_pre,1024,Fs);

%% Magnitude Response
```

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```
figure;
plot(f,20*log10(abs(H1)),'b','LineWidth',1.5);
hold on;
plot(f,20*log10(abs(H2)),'r--','LineWidth',1.5);
plot(f,20*log10(abs(H3)),'k-.','LineWidth',1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Magnitude Response Comparison');
legend('Impulse Invariance',...
    'Bilinear',...
    'Bilinear with Prewarping');
%% Phase Response
figure;
plot(f,unwrap(angle(H1))*180/pi,'b','LineWidth',1.5);
hold on;
plot(f,unwrap(angle(H2))*180/pi,'r--','LineWidth',1.5);
plot(f,unwrap(angle(H3))*180/pi,'k-.','LineWidth',1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (Degrees)');
title('Phase Response Comparison');
legend('Impulse Invariance',...
    'Bilinear',...
    'Bilinear with Prewarping');

%% Display Filter Coefficients
disp('Impulse Invariance Numerator Coefficients');
disp(b_imp);
disp('Impulse Invariance Denominator Coefficients');
disp(a_imp);
```

```
disp('Bilinear Numerator Coefficients');

disp(b_blt);

disp('Bilinear Denominator Coefficients');

disp(a_blt);

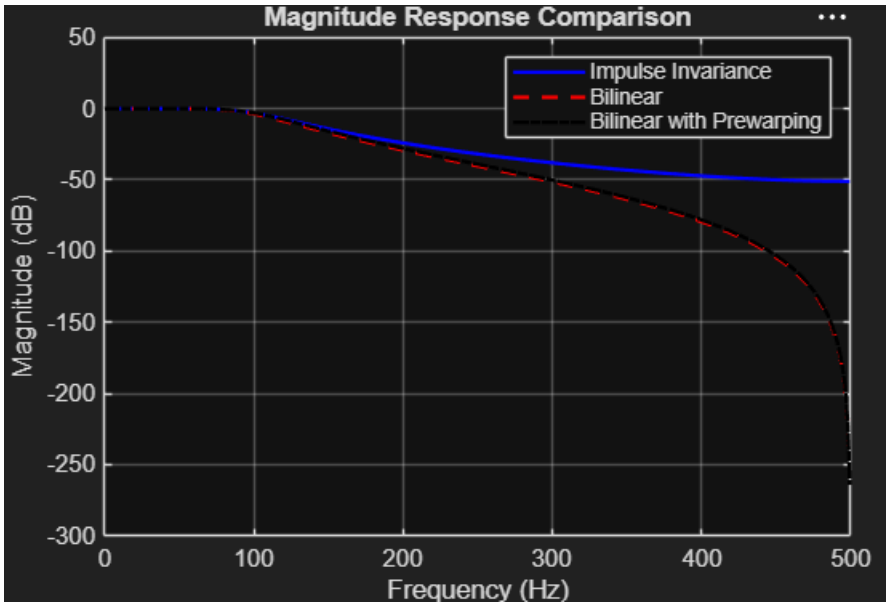
disp('Bilinear with Prewarping Numerator Coefficients');

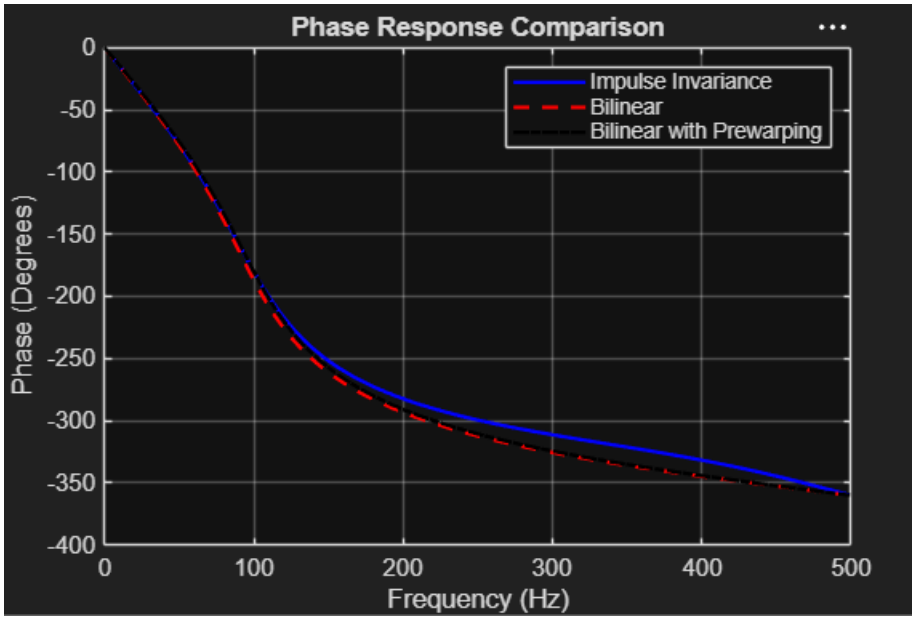
disp(b_pre);

disp('Bilinear with Prewarping Denominator Coefficients');

disp(a_pre);
```

OUTPUT





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```
Impulse Invariance Numerator Coefficients
   -0.0000    0.0169    0.0442    0.0075         0

Impulse Invariance Denominator Coefficients
    1.0000   -2.4020    2.3608   -1.0839    0.1936

Bilinear Numerator Coefficients
    0.0043    0.0173    0.0260    0.0173    0.0043

Bilinear Denominator Coefficients
    1.0000   -2.4196    2.3945   -1.1034    0.1978

Bilinear with Prewarping Numerator Coefficients
    0.0048    0.0193    0.0289    0.0193    0.0048

Bilinear with Prewarping Denominator Coefficients
    1.0000   -2.3695    2.3140   -1.0547    0.1874
```

Result:

A fourth-order analog Butterworth low-pass filter was successfully transformed into digital IIR filters using both **Impulse Invariance** and **Bilinear Transformation** methods. The frequency responses and filter coefficients were compared, showing that the bilinear transformation introduces frequency warping, while **prewarping** effectively compensates for this effect and provides a more accurate cutoff frequency.